WhatsApp Chat Analysis with Statistical Insights, Visualization, and Sentiment Analysis

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github.com/rohailahmad-code/Whatsapp-Chat-Analyzer

1. Project Overview

This project aims to extract meaningful insights from WhatsApp chat data using a combination of **statistical analysis**, **visualization**, and **sentiment analysis** techniques. Implemented in Python with a **Streamlit** interface, the project offers a user-friendly tool to understand communication patterns, user engagement, and emotional tone in chat conversations. The analyzer processes raw exported WhatsApp `.txt` files, applies a series of NLP and statistical operations, and presents results through interactive charts and dashboards.

2. Objectives

- Analyze chat data to identify communication and activity patterns
- Visualize message frequency, word usage, and emoji trends
- Perform sentiment analysis to detect emotional tone of conversations
- Build an interactive web application for easy data interpretation

3. Research Question

> *"How can statistical analysis, visualization, and sentiment analysis of WhatsApp chat data enhance our understanding of communication patterns and user engagement dynamics?"*

4. Literature Review Highlights

- **Previous Work:** Focused on Twitter sentiment, Facebook interactions
- **Identified Gaps:**
  - Lack of integrated tools combining stats, visualizations, and sentiment
  - Limited focus on media/emoji analysis and informal text sentiment
- **Tools Used in Literature:** NLTK, TextBlob, VADER for sentiment analysis

5. Methodology

1. **Preprocessing:** Clean and structure exported WhatsApp chat text files — parse timestamps, sender names, and message content; remove system messages; normalize encoding.
2. **Statistical Analysis:**
   - Message counts (total, per user, per day/month)
   - User activity breakdown (most active hours, busiest days)

- Emoji / media usage frequency

3. **Visualization:**
   - Bar graphs and line plots using Matplotlib & Seaborn
   - WordCloud for most used terms
   - Emoji frequency heatmaps

4. **Sentiment Analysis:** TextBlob and VADER to score polarity and subjectivity across individual messages and aggregated conversation windows.

6. Tools & Technologies

Category	Tool / Library	Purpose
Language	Python 3.x	Core implementation language
Web Interface	Streamlit	Interactive dashboard & UI
Visualization	Matplotlib, Seaborn	Charts, bar graphs, line plots
Visualization	WordCloud	Word frequency visualization
NLP	NLTK	Tokenization, stopword removal
Sentiment	TextBlob	Polarity & subjectivity scoring
Sentiment	VADER	Compound sentiment scoring for informal text
Data Processing	Pandas, Regex	Parsing, cleaning, structuring chat data

7. Contribution

Comprehensive Integration
Unites statistical analysis, visual exploration, and NLP techniques into a single pipeline.

User-Friendly
Streamlit app provides intuitive, no-code access to insights from any exported WhatsApp chat.

Engagement Insights
Focus on media sharing patterns, emoji usage behavior, and conversational sentiment tone.

8. Experimental Setup

1. Export chat from WhatsApp as `.txt` (without media, or with media for attachment tracking)
2. Upload it to the Streamlit interface via the file uploader widget
3. Automated analysis and visualization display — no additional configuration required

4	7+	Real-time
Analysis Modules	Visualization Types	Streamlit Dashboard

9. Future Enhancements

→ Multilingual sentiment support for non-English chats

→ Group vs. personal chat comparison analytics

→ BERT-based advanced sentiment classification models

→ Topic modeling (LDA) for conversation clustering

→ Reply-chain and thread analysis between participants

→ Time-series forecasting of chat activity trends

Note: The analysis is based on user-exported data and is intended for educational and analytical purposes only. No data is stored or transmitted beyond the local session.

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